

CYBERLAB COMPANION

Complete User Guide & Feature Walkthrough

AI-Powered Pentest & CTF Companion
v1.0 // All Features Covered

Navigate every screen. Verify every button.

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1. Getting Started — Overview & Launch

CyberLab Companion is a desktop Electron application designed to be your all-in-one assistant during Hack The Box, TryHackMe, and real-world pentesting engagements. It combines an AI chat interface, session tracking, tool libraries, command building, reverse shell generation, encoding utilities, cheatsheets, and per-phase finding workspaces — all in one dark-themed interface.

Launching the App

1	Open a terminal in the project folder Navigate to the CyberLab Companion directory on your machine.
2	Install dependencies (first run only) Run: <code>npm install</code>
3	Start the app Run: <code>npm start</code> — the main window opens.
4	Dashboard appears The Dashboard is the default landing screen. No session is auto-created.

Layout Overview

Feature	Description
Left Sidebar	Icon-based navigation. Click any icon to switch screens. Active screen is highlighted.
Main Area	Changes based on the selected screen (Dashboard, Lab, Commands, etc.)
Top Bar	Shows active session name, target info, and global action buttons.
Right Panel (Lab screen)	Tool library, notes, and bookmarks — context-sensitive to the active session.

Important: The app must be launched via `npm start` from the project directory. Do not open `index.html` directly in a browser — it requires Electron APIs.

2. Dashboard — Your Command Centre

The Dashboard is the first screen displayed when you launch CyberLab Companion. It gives you an at-a-glance overview of your current engagement status and provides quick access to the most-used features.

Dashboard Sections

Feature	Description
Hero Banner	Displays your operator name/handle and a greeting. Set your name in Settings.
Stats Row	Four stat cards: Active Sessions, Flags Captured, Machines Owned, Total Notes. Updated live as you work.
Recent Sessions	List of your last sessions. Click any entry to instantly resume that session in the Lab.
Quick Actions	Four action buttons — New Session, Recon Tool, Commands Library, Settings.
Tip of the Day	A rotating pentest tip, changes daily.
Phase Grid	Six cards (Exploits, Enum, PrivEsc, Post-Ex, Loot) — click to jump directly to a phase workspace.

Testing the Dashboard

1	Verify Stat Cards Check that the four stat cards display (they may show 0 if no sessions exist yet).
2	Click New Session Should open the New Session modal. Fill in a target name and click Start — you'll be taken to the Lab.
3	Return to Dashboard Click the house/dashboard icon in the left sidebar to go back.
4	Click a Phase Card In the Phase Grid, click Exploits — it should navigate to the Exploits phase workspace.

5**Verify Recent Sessions**

After creating a session, return to the Dashboard — it should appear in Recent Sessions.

6**Click a Recent Session**

Click it — you should be taken back to the Lab with that session active.

3. Lab — Session Management

The Lab is the core working screen. Each "tab" in the Lab represents a separate hacking session (one per target machine). You can have multiple sessions open simultaneously.

Starting a Session

1	Open the New Session modal Click the + button in the tab bar, or the "New Session" button on the Dashboard.
2	Fill in target details Enter a session name (e.g. "Lame"), target IP, hostname, OS platform, and difficulty.
3	Click Start The session tab is created and the Lab loads with your target pre-filled.

Session Tab Bar

The tab bar runs along the top of the Lab area. Each tab shows the session name. Click a tab to switch to it. The active tab is highlighted with the accent colour. Tabs can be closed with the X button on each tab.

Session Info Panel

Feature	Description
Target IP	The machine's IP address. Populated from the New Session form. Editable inline.
Hostname	The machine's hostname (e.g. lame.htb). Editable inline.
Platform	OS/platform dropdown — Linux, Windows, macOS, Web, Android, Other.
Difficulty	HTB difficulty: Easy, Medium, Hard, Insane.
Status Badge	Current status of the session (Active, Owned, etc.).

Testing the Lab

1	Create a new session Click +, fill in "Test Target" as name, IP "10.10.10.1", OS "Linux".
2	Verify session info shows The right panel should display the target IP and OS you entered.
3	Send a chat message Type "What is nmap used for?" in the message input and press Enter or click Send.
4	Verify AI responds The AI should reply in the chat panel within a few seconds.
5	Verify tab switching Create a second session and switch between tabs — each should retain its own chat history.
6	Close a tab Click the X on a session tab — it should close and switch to the next tab.

4. Chat Panel — AI Assistant

The Chat panel is the left portion of the Lab screen. It connects to an AI model and is your primary tool for getting help with exploitation, enumeration, privilege escalation, and any CTF challenge.

Sending Messages

1	Click the message input The text area at the bottom of the Chat panel.
2	Type your question or command You can ask anything: "How do I exploit SMBv1?", "Explain this error:", etc.
3	Press Enter or click Send The message is sent to the AI. A loading indicator shows while waiting for a response.
4	Read the response The AI reply appears in the chat bubble above your message.

Chat Features

Feature	Description
Session Context	The AI is aware of your current target (IP, OS, hostname) and uses it in responses.
Auto-Findings	The AI automatically extracts findings (IPs, ports, credentials, flags) from responses and saves them.
Copy Response	Each AI message has a copy button to copy the full response text.
Send to Chat	Tool cards in the right panel and cheatsheet items have a "Send to Chat" button.
Context Pills	Quick-action pills above the input (Enumerate, Exploit, PrivEsc, etc.) pre-fill suggested prompts.

TIP: If the chat input is not clickable, scroll down to ensure the input area is fully visible.

5. Tools Panel — Right-Side Library

The right side of the Lab screen contains a tool library. You can browse, filter, and search tools relevant to your current phase of the engagement.

Filter Tabs

At the top of the tools panel are filter buttons: **All**, **Recon**, **Enum**, **Exploit**, **PrivEsc**, **Post**. Click any filter to narrow the displayed tools.

1	Click "Recon" filter The tool list should update to show only recon tools (nmap, gobuster, etc.).
2	Click "Exploit" filter Should show exploit-related tools.
3	Click "All" Returns to showing all tools.
4	Use the search bar Type a tool name (e.g. "nmap") to search within the current filter.

Using a Tool Card

Feature	Description
Tool Name & Icon	Displayed at the top of each tool card.
Description	A brief explanation of what the tool does.
Copy Command	Copies the tool's default command to clipboard.
Send to Chat	Pastes the command into the chat input so you can discuss it with the AI.
Paste to Terminal	Copies the command to clipboard and shows a paste notification.

6. Notes & Bookmarks

Notes and Bookmarks are accessible both from the left sidebar (as standalone screens) and from the bottom-right panel within the Lab screen. Notes are saved per-session when a session is active, or globally when no session is open.

Notes Screen

1	Open Notes Click the Notes icon in the left sidebar (or switch to the Notes tab in the Lab panel).
2	Add a note Type in the input field at the top of the Notes panel and click the + button or press Enter.
3	Bookmark a note Click the star (★) icon on any note to mark it as bookmarked.
4	Copy a note Click the copy icon on a note to copy its text to the clipboard.
5	Delete a note Click the × icon on a note to delete it permanently.

Bookmarks Screen

The Bookmarks screen shows all notes that have been starred. Click the Bookmarks icon in the left sidebar to access it. To remove a bookmark, click the × button on the bookmarked note — this removes the bookmark flag (the note is not deleted).

Notes Tab in Lab Panel

In the Lab screen, the bottom-right panel has two tabs: **Notes** and **Bookmarks**. Click either tab to switch between them within the Lab context. Notes added here are saved to the currently active session.

Verify: Click the "Notes" tab and "Bookmarks" tab in the Lab panel bottom-right — both should switch correctly.

7. Commands Screen

The Commands screen is your all-in-one toolkit for building penetration testing commands, generating reverse shells, encoding/decoding data, and referencing cheatsheets. Access it via the terminal/commands icon in the left sidebar.

At the top of the Commands screen are five tab buttons: **Command Builder**, **Reverse Shells**, **Encoder**, **Cheatsheets**, and **References**. Click any tab to switch to that sub-tool. The active tab is highlighted.

7.1 Command Builder

The Command Builder lets you construct tool commands by selecting a category, picking a tool, and filling in parameters. It then generates the full command for you.

1	Select a category Click the Category dropdown — choose from Recon, Enumeration, Exploitation, PrivEsc, Post-Exploitation, etc.
2	Select a tool The Tool dropdown updates to show tools in the chosen category. Select your tool.
3	Fill in parameters Parameter fields appear for that tool (target IP, port, wordlist, etc.). Fill in the required fields (marked with *).
4	Click Build Command Click the "Build Command" button. The generated command appears in the output box below.
5	Copy the command Click the copy button next to the output to copy it to your clipboard.

Note: If a session is active with a target IP set, the target parameter is pre-filled automatically.

7.2 Reverse Shells

The Reverse Shells tab generates ready-to-use reverse shell payloads in over a dozen languages and supports multiple encoding options.

1	Select language Choose from the Language dropdown: Bash, Python, PHP, Perl, Ruby, PowerShell, Java, C, Go, Lua, and more.
2	Enter your IP Type your listener/tun0 IP address in the "Your IP" field.
3	Enter port Enter the listener port (default 4444). Pick any unused port.
4	Choose encoding Optionally select an encoding: Raw, Base64, URL-encoded, or Unicode.
5	View outputs Three output sections appear: Payload, Listener Command, Shell Stabilisation commands.
6	Copy payload Click the copy button next to the Payload section to copy just the shell payload.
7	Copy listener Copy the nc -lvnp command for your listener terminal.

7.3 Encoder / Decoder

The Encoder tab handles common encoding and hashing operations needed during CTF challenges.

Feature	Description
Base64 Encode / Decode	Standard Base64 encoding and decoding.
URL Encode / Decode	Percent-encoding for URLs.
HTML Encode / Decode	HTML entity encoding and decoding.
Hex Encode / Decode	Converts text to/from hexadecimal.
ROT13	Classic Caesar cipher rotation.
Text to Binary / Binary to Text	Converts text to binary representation and back.
MD5 / SHA1 / SHA256 Hash	Generates cryptographic hashes of the input.

JWT Decode		Decodes a JSON Web Token (header and payload — does not verify signature).
1	Enter input text	Type or paste the text/data you want to process in the top input area.
2	Select operation	Choose an operation from the dropdown.
3	Click Run	The result appears in the output area below.
4	Swap	Click the Swap button to move the output back to the input for chained operations.

7.4 Cheatsheets

The Cheatsheets tab provides a curated reference of commands organised by topic. Select a topic from the dropdown to browse its commands.

1	Select a topic	Use the Cheatsheet dropdown to pick a topic (e.g. Linux PrivEsc, SMB Enum, Web Attacks).
2	Browse sections	Commands are grouped into logical sections within each topic.
3	Copy a command	Click the copy icon next to any command to copy it to clipboard.
4	Discuss with AI	Click the chat icon next to a command to send it to the AI chat for explanation.

7.5 References

The References tab provides clickable cards linking to key external resources: HackTricks, GTFOBins, ExploitDB, LOLBAS, CyberChef, RevShells, PayloadsAllTheThings, and more. Click any card to open the URL in your default browser.

1	Switch to References tab Click the "References" tab button at the top of the Commands screen.
2	Click a reference card Any card (e.g. HackTricks) should open the URL in your browser.

8. Recon Screen

The Recon screen is a structured form for capturing scan results and getting AI-powered analysis. It is designed to be used alongside nmap, gobuster, nikto, and other enumeration tools.

Recon Form Fields

Feature	Description
Target IP	The machine's IP address. Auto-filled from the active session if one is open.
Hostname	The machine's hostname (e.g. legacy.htb).
Operating System	OS dropdown: Linux, Windows, macOS, Unknown.
Open Ports	List of discovered open ports and services (e.g. 22/tcp ssh, 80/tcp http).
Credentials Found	Any credentials discovered during recon.
Raw Scan Output	Paste your full nmap, gobuster, nikto, or other tool output here for AI analysis.

Using the Recon Screen

1	Open Recon Click the Recon icon in the left sidebar.
2	Fill in target details Enter the IP, hostname, and OS.
3	Paste scan output Copy your nmap -sV output and paste it into the Raw Scan Output field.
4	Click Analyse The AI analyses the output and provides attack vectors, CVEs, and next steps. Check the Chat tab for results.

5**Click Apply to Session**

Saves the IP, hostname, OS, ports, and credentials to the active session and returns you to the Lab.

Verify: After clicking Apply to Session, the target IP field in the Lab session info panel should update to the value you entered in the Recon form.

9. Phase Workspaces

Phase Workspaces are dedicated screens for each stage of a pentest or CTF engagement. They provide a two-column layout: the left side tracks your findings for that phase, and the right side provides a filtered tool library to assist with the phase.

Access any phase workspace from the left sidebar icons or from the Dashboard Phase Grid. The five phases are: **Exploits**, **Enumeration**, **Privilege Escalation**, **Post-Exploitation**, and **Loot**.

Phase Workspace Layout

Feature	Description
Left Column — Findings	Your notes and findings for this phase. Add, tag, copy, and delete findings.
Right Column — Tool Library	Tools relevant to this phase. Filter by keyword and click tools to add them to chat.

Adding a Finding

1	Type your finding Click the input field at the top of the Findings column and type your finding text.
2	Select a status tag Choose a status from the dropdown: Worked, Done, Failed, Tried, Found, Interesting, Credential, Hash, Flag, Key.
3	Click Save / press Enter The finding is saved and appears in the list below with a coloured status badge.
4	Copy a finding Click the copy icon on any finding row to copy the text.
5	Chat about a finding Click the chat icon on a finding to send it to the AI for analysis.
6	Delete a finding Click the × icon on a finding to remove it.

Using the Tool Library (right column)

1	Search for a tool Type in the search box above the library to filter by name or description.
2	Click a tool card Click the "Add to Chat" button on a library card to paste the command into the chat input.

9.1 Exploits

The Exploits workspace stores all your exploit attempts and results. Use it to track which exploits you've tried, which worked, and which failed. The tool library shows exploit frameworks and scripts.

9.2 Enumeration

The Enumeration workspace tracks services, endpoints, usernames, and other data gathered during the enumeration phase. Tools include directory bruteforcers, SMB enumeration, and more.

9.3 Privilege Escalation

The PrivEsc workspace stores escalation paths, SUID binaries, weak configs, kernel exploits, and any credentials that allowed privilege escalation. Tools include linPEAS, winPEAS, and GTFOBins helpers.

9.4 Post-Exploitation

The Post-Exploitation workspace tracks lateral movement, persistence mechanisms, and data gathered after initial access. Tools include mimikatz, secretdump, and session management utilities.

9.5 Loot

The Loot workspace is slightly different from the other phases. The right column contains static loot command cards rather than a dynamic library. These cards provide ready-made commands for extracting credentials, hashes, SSH keys, and sensitive data:

Feature	Description
Hash Cracking	john and hashcat commands for cracking recovered hashes.
Secretsdump	impacket-secretsdump command for dumping Windows credentials.
Mimikatz	Common mimikatz commands for credential extraction.
SSH Keys	Locating and using id_rsa private keys found on target machines.
/etc/shadow	Reading and cracking Linux shadow file entries.
Environment Variables	Enumerating environment variables for secrets and API keys.

10. Settings

Access Settings by clicking the gear icon in the left sidebar. The Settings screen is scrollable — scroll down to see all sections. Click the Back button to return to the Dashboard.

Settings Sections

Feature	Description
Theme Picker	Five themes: Hacker (default green), Cyberpunk (pink/purple), Nord (blue), Dracula (purple), Terminal (classic). Click any theme card to apply immediately.
Operator Profile	Set your operator name/handle. This name appears on the Dashboard greeting and in session logs.
AI Model	Select the AI model tier. Different tiers offer different response speeds and capabilities.
Manage Tools	Open the Manage Tools modal to add custom tool cards to the right-panel library. Add name, description, category, and command.
Notifications	Toggle sound effects and toast notification visibility on/off.
Reset Data	Clears all session data and resets the app to its initial state. Use with caution.

Changing the Theme

1	Open Settings Click the gear icon in the sidebar.
2	Scroll to Theme Picker The theme cards are near the top of the settings page.
3	Click a theme card For example, click "Cyberpunk". The entire app should instantly change colour scheme.
4	Verify colours updated The sidebar, panels, buttons, and text should all reflect the new theme.

5**Switch back to Hacker**

Click the Hacker theme card to restore the default green theme.

Adding a Custom Tool

1**Open Settings**

Click the gear icon.

2**Click Manage Tools**

The Manage Tools button opens a modal.

3**Click Add Tool**

Opens an input form within the modal.

4**Fill in tool details**

Enter: Tool Name, Description, Category (recon/enum/exploit/privesc/post), and Command string.

5**Save the tool**

Click Save. The tool appears in the right-panel library when the matching filter is active.

11. Tips & Keyboard Shortcuts

The following tips will help you get the most out of CyberLab Companion during active engagements.

Workflow Tips

- Start each box by creating a new session from the Dashboard. Fill in all target details upfront — the AI and Command Builder use them.
 - Use the Recon screen to paste your initial nmap output. Let the AI suggest attack vectors before you dig in.
 - As you work, use the Phase Workspaces to track findings in context. Mark findings with the correct status tag (Worked, Failed, Found, etc.) to keep a clear picture of progress.
 - Use the Command Builder for any tool you're less familiar with. It generates correctly-formatted commands with your target pre-filled.
 - Before privilege escalation, switch to the PrivEsc workspace and run linPEAS or winPEAS. Paste interesting output into the Findings column for reference.
 - When you capture a flag, add it as a finding in the Loot workspace with the "Flag" status tag. It will be counted in your Dashboard stats.
 - The Encoder is handy for CTF challenges involving base64, URL encoding, or JWT tokens — no need to leave the app.
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Chat Tips

- Be specific in chat prompts. Instead of "help me", try: "I found SMB port 445 open on a Windows machine. What should I enumerate first?"
 - Paste error messages directly into chat. The AI will diagnose them and suggest fixes.
 - Use the context pills above the message input for common pentest queries — they pre-fill a structured prompt.
 - The AI remembers your target IP, OS, and hostname within the session — you don't need to re-state it each time.
-

Reverse Shell Tips

- Always set up your listener before delivering the payload: `nc -lvnp 4444`
- If a raw bash shell is blocked, try Base64-encoded or URL-encoded variants from the Reverse Shell generator.
- After getting a shell, use the Stabilisation commands shown in the Reverse Shells output to get a fully interactive TTY.

12. Troubleshooting

Feature	Description
Buttons not responding	Refresh the app (Ctrl+R or Cmd+R). If the issue persists, check the DevTools console (Ctrl+Shift+I) for JavaScript errors.
Chat input not clickable	Scroll down in the Lab screen to ensure the input area is fully visible and not obscured by another element.
Settings screen does not scroll	This was a known display bug (now fixed). If you see this, ensure you have the latest renderer.js built from all three parts.
Category dropdown shows "Select category..." after picking	This was a wiring bug in wireCommandsScreen (now fixed with <code>_wired</code> guards). Rebuild <code>renderer.js</code> and reload the app.
Command Builder output is wrong or empty	Ensure a tool is selected AND all required parameters are filled. Required fields are marked with <code>*</code> .
AI not responding	Check your API key is configured correctly in the Electron main process. The AI requires a valid API key to function.
Session data not saving	CyberLab Companion saves data to <code>localStorage</code> via Electron. If data is lost after restart, check the Electron <code>userData</code> path and ensure write permissions.
Theme not applying	Click the theme card once firmly. The theme system sets a <code>data-theme</code> attribute on the document body — open DevTools to verify it changed.
Tool library empty	Ensure the filter is set to "All" or a category that has tools. Custom tools appear only when the matching category filter is active.
App won't start	Run <code>npm install</code> first, then <code>npm start</code> . Ensure Node.js <code>>= 16</code> and Electron are installed. Check the terminal for error output.

Rebuilding `renderer.js`

After any edits to the `renderer` source files, you must rebuild `renderer.js` by running the following command from the project root:

```
cat renderer-part1.js renderer-part2.js renderer-part3.js > renderer.js
```

Verify syntax with:

```
node --check renderer.js
```

CyberLab Companion is an ongoing project. If you encounter bugs not covered here, check the browser DevTools console for error messages and ensure you are running the latest rebuilt `renderer.js`.

Happy Hacking. Good luck on the box.